Ch 5.1.5: k-fold Cross-Validation for Classification

Lecture 15 - CMSE 381

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Mon, Oct 7, 2024

Announcements

Last time:

- k-fold CV

This lecture:

- CV for classification

Announcements:

- Homework #4 is posted, Due Wednesday
- Grades

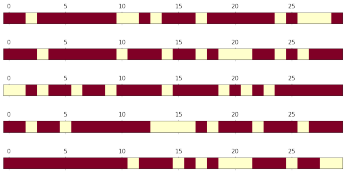
Percent	Convert
≥ 90%	4.0
≥ 85%	3.5
≥ 80%	3
≥ 75%	2.5
≥ 70%	2
≥ 65%	1.5
≥ 60%	1
< 60%	0

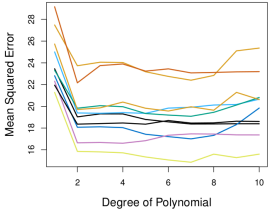
Section 1

Last time

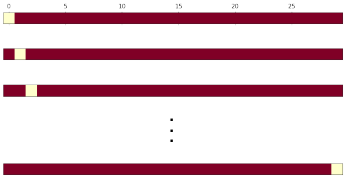
Approximations of Test Error

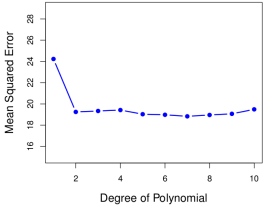
Validation Set



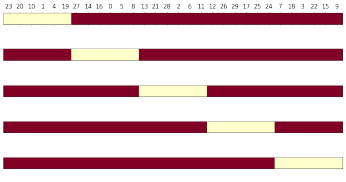


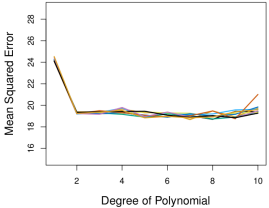
LOOCV





K-fold CV



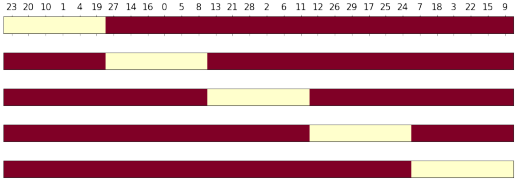


Definition of k-fold CV

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$

- Rinse and repeat



Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

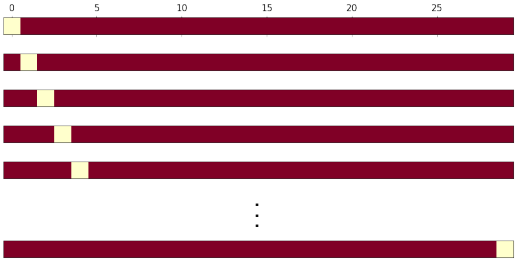
Section 2

CV for Classification

Setup: LOOCV

- Remove ith point (x_i, y_i) and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$

- Rinse and repeat



Return

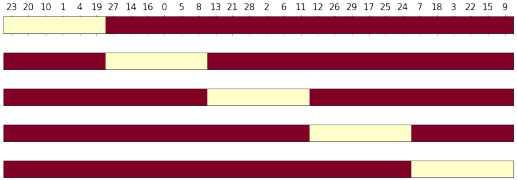
$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{Err}_i$$

Setup: k-fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \mathrm{I}(y_j \neq \hat{y}_j)$

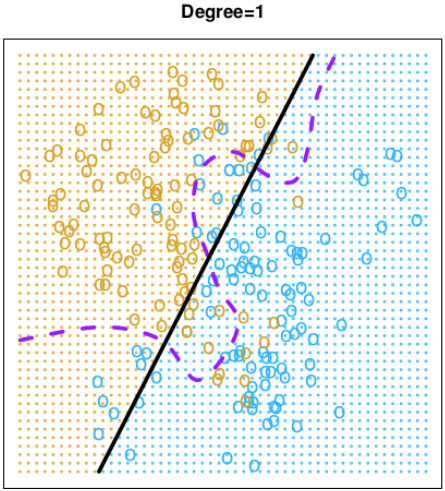
- Rinse and repeat



Return

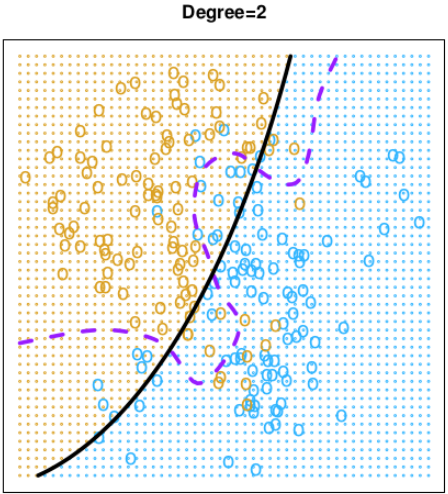
$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

Example on simulated data: Linear



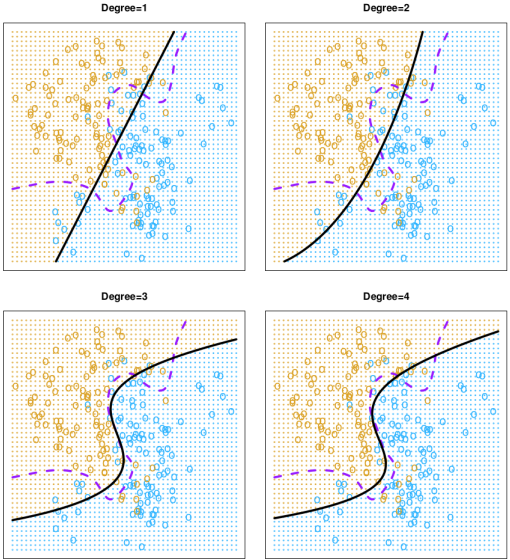
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
  - Error rate: 0.201
- No k-fold yet
- Error rate for logistic still high, so not great job yet
- So, up the degree and see what happens

Example on simulated data: Quadratic logistic regression



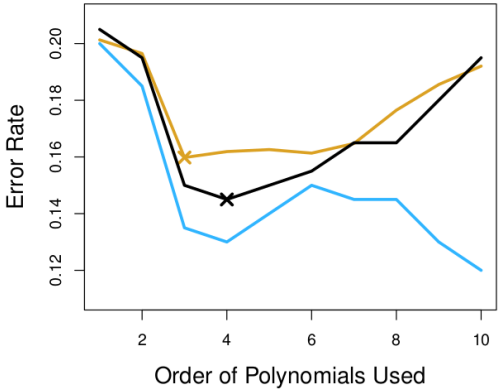
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2$
  - Error rate: 0.197
- No k-fold yet
- Error rate for logistic slight improvement, but still not great
- So, up the degree again and see what happens

Example on simulated data: all the polynomials!



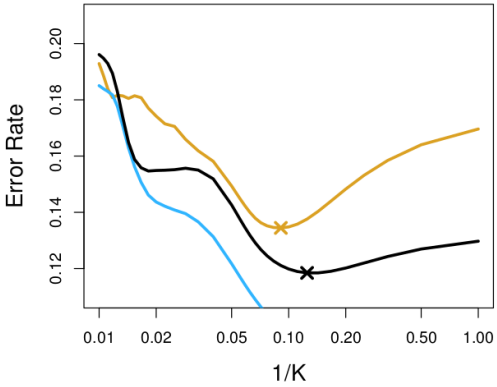
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - Deg 1 Error rate: 0.201
  - Deg 2 Error rate: 0.197
  - Deg 3 Error rate: 0.160
  - Deg 4 Error rate: 0.162
- Normally, we don't have the Bayes decision boundary
- How do you pick between models?

Decide degree based on CV



- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)
- Training error tends to decrease as the flexibility of the fit increases.
- Note decrease in blue isn't monotonic, but still going down overall
- test error displays a characteristic U-shape.
- The 10-fold CV provides a pretty good (but underestimate) approximation to the test error rate.
- minimum at 4, close to minimum of test curve at 3

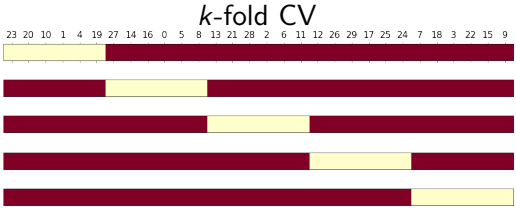
Similar game for KNN



- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)
- Note that using a different changing parameter
- similar idea to figure out the right choice of K.
- Minimm in black (10 fold CV) has $1/K$ value close to where brown does, so good choice

Coding - k-fold for penguin classification section

TL;DR



$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

Use $k = 5$ or 10 usually

k-fold CV for classification

$$\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$$

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i$$

Next time

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Resampling methods: Bootstrap	5.2	HW #4 Due Weds 10/9
17	Fri	10/11	Subset selection	6.1	
18	Mon	10/14	Shrinkage: Ridge	6.2.1	
19	Wed	10/16	Shrinkage: Lasso	6.2.2	
20	Fri	10/18	Dimension Reduction	6.3	HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		